MIL-T-43664B
9 February 1988
SUPERSEDING
MIL-T-43664A
20 November 1986

MILITARY SPECIFICATION

TAPE, TEXTILE, NYLON OR POLYESTER (FOR NAME TAPES)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* 1.1 Scope. This document covers two types of tape.

* 1.2 Classification. The tape shall be of the following types as specified (see 6.2):

Type I - Nylon
Type II - Polyester

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Documents. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8315

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SPECIFICATION

MILITARY

MIL-P-43334 - Packaging of Textile Webbing and Tape

STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.1.2 Other Government documents. The following other government documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

Rules and Regulations Under the Textile Fiber Products Identification Act

(Application for copies should be addressed to the Federal Trade Commission, Washington, DC 20580.)

2.2 Other publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of the solicitation.

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

Chromatic Transference Scale

(Application for copies should be addressed to the American Association of Textile Chemists and Colorists, P.O. Box 12215, Research Triangle Park, NC 27709-2215.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Standard sample. The dyed tape shall match the standard sample for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.2 Material.

3.2.1 Yarn.

* 3.2.1.1 Nylon yarn (type I). The yarn for type I tape shall not be subjected to any type of bleaching process. The yarn shall be a bright, high tenacity, light and heat resistant polyamide prepared from either hexamethylene diamine and adipic acid with a minimum melting point of 472°F or caprolactam with a minimum melting point of 430°F when tested as specified in 4.2.3.

* 3.2.1.2 Polyester yarn (type II). The yarn for type II tape shall be semi-dull or bright, regular or high tenacity fiber prepared from polyethylene glycol terephthalate with a minimum melting point of 472°F when tested as specified in 4.2.3.

* 3.2.2 Yarn number. The yarn number for the warp and filling, prior to dyeing, shall be as specified in 3.4.

3.3 Color. The color of the tape shall be as specified (see 6.2).

3.3.1 Dyeing. The tape shall be yarn or piece dyed and shall not be subjected to any type of bleaching process.

3.3.2 Color matching. The dyed tape shall match the standard sample when viewed under filtered tungsten lamps that approximate artificial daylight and that have a correlated color temperature of 7500 ± 200 K, with illumination of 100 ± 20 foot candles, and shall be a good match to the standard sample under incandescent lamplight at 2300 ± 200 K.

3.3.3 Colorfastness. The dyed tape shall show fastness to light, perspiration, and laundering (after 3 cycles) equal to or better than the standard sample or equal to or better than a rating of "good" when tested as specified in 4.2.3. The dyed tape shall show fastness to crocking equal to or better than the standard sample or shall have an AATCC Chromatic Transference Scale rating not lower than 3.5 when tested as specified in 4.2.3.

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3.4 Physical requirements. The tape shall conform to the require-ments specified in table I when tested as specified in 4.2.3.

* TABLE I. Physical requirements

Requirement	Type I (Nylon)	Type II (Polyester)
Width (inches)	1 ± 1/32	1-1/16 ± 1/32
Thickness (inches)	0.015 to 0.025	0.006 to 0.012
Weight (oz/yd), maximum	0.30	0.20
Warp ends (face and back), minimum	200	125
Picks per inch, minimum	33	80 1/
Yarn numbers:		
Warp (nominal)	210 (den)	170 (den)
Filling (nominal)	420 (den)	40/1

1/ Two picks per shed

* 3.5 Weave. The weave for both types of tape shall be plain weave 1-up and 1-down. The filling yarn for type I shall traverse the full width of the tape with one filling yarn per shed. The filling yarn for type II shall traverse the full width of the tape with two filling yarns per shed and shall be held at the edge by a catch cord yarn interlacing with the filling yarn.

3.6 Nonfibrous materials. The water-soluble and chloroform-soluble materials of the tape shall be no more than 2.0 percent when tested as specified in 4.2.3.

3.7 Dimensional stability. The shrinkage of the tape in the warp or lengthwise direction shall be no more than 5.0 percent when tested as specified in 4.2.3.

3.8 pH. The pH value of the water extract of the finished tape shall be no less than 5.0 nor more than 8.5 when tested as specified in 4.2.3.

3.9 Length and put-up. For Government procurement only, unless otherwise specified (see 6.2), the tape for type I shall be furnished in rolls of 72 yards +/- 1 yard. The tape for type II shall be furnished in rolls of 200 yards +/- 1 yard. No roll shall contain more than two pieces, and no piece shall be less than 5 yards. Ends of pieces may be butted and fastened with adhesive tape. Overlapping pieces and sewing is prohibited.

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3.10 Identification tickets. Each roll of tape shall have an identification ticket attached to the roll in accordance with MIL-W-43334.

3.11 Fiber identification. Each roll of tape shall be labeled or ticketed for fiber content in accordance with the Textile Fiber Products Identification Act.

3.12 Workmanship. The finished tape shall conform to the quality and grade of product established by this document. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirement in the document shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.2.2 End item examination.

4.2.2.1 Yard-by-yard examination. The required yardage from each roll shall be inspected on both sides for the visual defects listed in table III. All defects found shall be counted regardless of their proximity one to each other

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except where two or more defects represent a single local condition of the tape, in which case only the more serious defect shall be counted. A continuous defect shall be counted as one defect for each warpwise yard or fraction thereof in which it occurs. The sample unit shall be 1 yard. The inspection level shall be III and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 0.40 for major defects and 10.0 for total combined major and minor defects. The number of rolls from which the sample is to be selected shall be in accordance with table II. The sample yardage shall be apportioned equally among the selected rolls.

TABLE II. Sample size

Lot size in yards	Sample size (rolls)	Acceptance number 2/
Up to 1,200 1/	3	0
1,201 up to an including 3,200	5	0
3,201 up to and including 10,000	8	0
10,001 up to and including 35,000	13	0
35,001 up to and including 150,000	20	1
150,001 and over	32	2

1/ If a lot contains fewer than three rolls, each roll in the lot shall be examined.

2/ See 4.2.2.2 and 4.2.2.3.

TABLE III. Tape visual defects

Examine	Defect	Classification Major	Classification Minor
Abrasion mark	Resulting in rupture of yarns or in nap sufficient to obscure the identity of any yarns, exceeding 10 percent of width or 1 inch in length	X	
Broken or missing end	Two or more regardless of length or a single end exceeding 6 inches in length	X	
	Single end, 1/4 inch up to and including 6 inches		X
Broken or missing pick	Two or more regardless of length	X	

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TABLE III. Tape visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Coarse or light filing bar	Resulting in noticeable difference in stiffness of thickness of tape and extending for more than 1/4 inch in length direction	X	
	Resulting in noticeable difference in stiffness or thickness of tape and extending for 1/4 inch or less in the length direction		X
Twist or distortion	Will not lay flat upon application of manual pressure		X
Cut, hole, or tear	Any	X	
Edges	Frayed, slack, tight, or otherwise poorly constructed and exceeding 1/4 inch in length	X	
Scalloped edges	Indentation of 1/8 inch or more, regardless of length	X	
Floats or skips	Multiple, 1/2 inch or more in warp and filling directions	X	
	Single float or skip, over 1 inch	X	
	Multiple, less than 1/2 inch in combined warp and filling directions		X
	Single float or skip over 1/2 inch but not exceeding 1 inch if in the warp		X
	Single float or skip over 1/4 inch but not exceeding 1 inch if in the filling		X
Hitchback crack	Opening between adjoining picks or warpwise tension area over part of the width resulting in light and heavy places 1/		X
Jerked-in filling, slough-off, slug	More than twice the thickness of the normal yarn		X
Kinks	More than three kinks in any 9 linear inches	X	

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TABLE III. Tape visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Knots	More than two knots in any 9 linear inches	X	
	Single knot with untrimmed ends extending more than 1/16 inch from the surface of tape		X
Mispick, double pick	Two or more across the full width	X	
	Single across the full width		X
Slack end	Two or more in the same area, jerked in between picks, or forming clearly visible loops on surface of tape 1/	X	
	Single jerked in between picks or forming clearly visible loops on surface of tape		X
Catch cord missing (applicable to shuttleless looms only)	Any	X	
Shift mark	Yarn slippage resulting in open place, clearly visible heavy and light yarn density 1/		X
Slub or slug, gout	More than twice the thickness of the yarn		X
Smash	Any	X	
Spot, stain, or streak	Any 1/		X
Tight end	Any 1/		X
Wrong draw	Extending for more than 9 inches		X
Width	Beyond specified tolerances		X
Uneven width	Tight or loose filling resulting in variations of ± 1/16 inch in width	X	

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1/ Clearly visible at normal inspection distance (approximately 3 feet).

4.2.2.2 Overall examination. The tape shall be examined for the defects listed below. Each defect listed shall be counted not more than once in each roll examined. The sample shall consist of the applicable number of rolls indicated in table II. The lot shall be rejected if the total number of defects in the sample exceeds the applicable acceptance number specified in table II.

Defects

Objectionable odor
Unclean throughout
Uneven weaving throughout
Off shade, not within established tolerance
Not labeled in accordance with Textile Fiber Products Identification Act

4.2.2.3 Length examination. Each roll of tape used in the overall examination shall be examined for the defects listed below. If the total number of defects in the sample rolls exceeds the applicable acceptance number specified in table II or if the total of the actual gross lengths of the sample rolls is less than the total of the gross lengths marked on the roll tickets, the lot shall be rejected.

Defects

Gross length less than specified minimum length or more than specified maximum length
Gross length more than 2 yards less than gross length marked on ticket
Any roll containing more than two pieces
Any piece less than 5 yards in length

4.2.3 End item testing. The tape shall be tested for the characteristics listed in table IV. The methods of testing specified in FED-STD-191, wherever applicable, and as listed in table IV shall be followed. The physical and chemical values specified in section 3 apply to the results of the determinations made on a sample unit for test purposes as specified in the applicable test methods. All test reports shall contain the individual values utilized in expressing the final result. The sample unit shall be 6 yards of tape. The sample size shall be as follows:

Lot size (yards)	Sample size (sample units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

The lot shall be unacceptable if one or more sample units fail to meet any requirement specified.

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* TABLE IV. End item tests

Characteristic	Requirement paragraph	Test method
Material (yarn):		
Nylon:		
Material identification	3.2.1.1	1530 1/
Melting point	3.2.1.1	1534 1/
Yarn number	3.2.2	4021 1/
Polyester:		
Material identification	3.2.1.2	1600 1/
Melting point	3.2.1.2	1534 1/
Yarn number	3.2.2	4021 1/
Colorfastness to:		
Light	3.3.3	5660
Perspiration	3.3.3	5680
Crocking	3.3.3	5651
Laundering (after 3 cycles)	3.3.3	5610 2/
Thickness	3.4	5030
Weight	3.4	5041
Warp ends (face and back)	3.4	5050
Picks per inch	3.4	5050
Weave	3.5	Visual
Nonfibrous material	3.6	2611
Dimensional stability	3.7	5550
pH	3.8	2811

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirement.

2/ The specimens must be dried after each of the 3 laundering cycles. On the color transfer cloth, the stain on the nylon, cotton, polyester, and wool fibers shall be evaluated.

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4.2.4 Packaging inspection. The inspection shall be in accordance with the quality assurance provisions of MIL-P-43334.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial, as specified (see 6.2).

5.1.1 Levels A and Commercial. Tape, put up as specified, shall be preserved in accordance with the applicable requirements of MIL-P-43334.

5.2 Packing. Packing shall be level A, B, or Commercial, as specified (see 6.2).

5.2.1 Levels A, B, and Commercial. The tape shall be packed in accordance with the applicable requirements of MIL-P-43334.

5.3 Marking. In addition to any special marking required by the contract or purchase order, shipments shall be marked in accordance with MIL-P-43334.

6. NOTES

6.1 Intended use. The tape is intended for use as name tapes for uniforms and field clothing.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. Type required (see 1.2).
c. Color required (see 3.3).
d. Length of roll if other than specified (see 3.9).
e. Selection of applicable levels of preservation and packing (see 5.1 and 5.2).

6.3 Standard sample. For access to samples, address the contracting activity issuing the invitation for bids.

6.4 Olive Drab 7. The standard sample dyed Olive Drab-7 for type I was prepared and skein dyed as follows: The greige tape was scoured with 0.2 percent nonionic detergent of the ethylene oxide condensate type and 2 percent ammonium hydroxide at 180°F. The dyeing was accomplished by the following formulation:

Acid Yellow 25, C.I. 18835
Acid Orange 128
Special Nylon Red B
Acid Blue 40, C.I. 62125

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* 6.5 Light tan 389. A suggested but not mandatory dye formulation for light tan on nylon is as follows:

Acid Blue 25
Nylanthrene Orange B-GN
Nylanthrene Red 2B

* 6.6 Camouflage Green 483. A suggested but not mandatory dye formulation for Camouflage Green 483 on nylon is as follows:

Lanaset Yellow 2R
Lanaset Brown B
Lanaset Blue 5G

* 6.7 Subject term (key word) listing.

Name tapes
Nylon tape
Polyester tape

6.8 Changes from previous issue. The margins of this document are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written, irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

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Review activities:

Army - MD
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Preparing activity:

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-43664B	Tape, Textile, Nylon or Polyester (For Name Tapes)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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